

Exercise 1.15 Let $A : \mathbb{R} \rightarrow \mathbb{R}^{m \times n}$, where $m \geq n$. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ as $f(t) = \det(A(t)^T A(t))^{\frac{1}{2k}}$ for some $k \in \mathbb{N}$.

1. Show that

$$\frac{df(t)}{dt} = \frac{f(t)}{k} \operatorname{tr} \left(A(t)^\dagger \frac{dA(t)}{dt} \right).$$

2. Suppose $A(t)$ has full rank and let $A(t) = QR$ be a reduced QR factorization of $A(t)$, i.e., $Q \in \mathbb{R}^{m \times n}$ and $R \in \mathbb{R}^{n \times n}$. Show that

$$\frac{df(t)}{dt} = \frac{1}{k} \left| \prod_{i=1}^n R_{ii} \right|^{1/k} \operatorname{tr} \left(R^{-1} Q^T \frac{dA(t)}{dt} \right).$$

Exercise 1.16 Recall that given a matrix $A \in \mathbb{R}^{m \times n}$, the projection of a point $x \in \mathbb{R}^m$ onto $\mathcal{N}(A^T)$ may be expressed as Px where $P = I - AA^\dagger$ is a projection matrix. If we assuming that $\operatorname{rank}(A) = n$, then P can be expressed as

$$P = I - A(A^T A)^{-1} A^T.$$

Now, suppose A is a function of a scalar parameter t .

1. Show that

$$\frac{dP}{dt} = -P \frac{dA}{dt} A^\dagger - (A^\dagger)^T \frac{dA^T}{dt} P.$$

2. Let $A = Q_1 R_1$ be a reduced QR decomposition of A (i.e., $Q_1 \in \mathbb{R}^{m \times n}$ and $R_1 \in \mathbb{R}^{n \times n}$). Show that a projection Px and the derivative dP/dt can be evaluated efficiently using such a QR decomposition without explicitly computing $A^\dagger = (A^T A)^{-1} A^T$.

Exercise 1.17 Consider the equation

$$Y = \mathcal{O}X + \mathcal{T}U$$

where $Y \in \mathbb{R}^{p \times m}$, $\mathcal{O} \in \mathbb{R}^{m \times n}$, $X \in \mathbb{R}^{n \times p}$, $\mathcal{T} \in \mathbb{R}^{p \times p}$ and $U \in \mathbb{R}^{m \times p}$.

1. Let $\Pi^\perp = I - U^T(UU^T)^{-1}U$ and show that

$$Y\Pi^\perp = \mathcal{O}X\Pi^\perp$$